

GSoC 2026 Project Proposal: Project PrediCT

Physio-Twin: A Physics-Informed Stochastic Framework for Generative Coronary Calcium Synthesis

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Difficulty: Advanced (350 Hours)

1 Abstract

Automated Coronary Artery Calcium (CAC) scoring is often limited by the scarcity of diverse clinical datasets, particularly for rare phenotypes and high Agatston scores. Current data augmentation techniques rely on heuristic rules that lack physiological realism. **This project re-frames CAC generation as a stochastic inverse problem governed by latent hemodynamic dynamics rather than a purely image synthesis task.** By integrating a validated, failure-aware atlas registration framework with **Latent Stochastic Differential Equations (SDEs)** and **Reduced-Order Physics-Informed Neural Networks (PINNs)**, the system synthesizes anatomically plausible calcium distributions. This approach shifts medical data augmentation from visual mimicry to physiological simulation, providing a high-fidelity engine for training robust clinical models.

2 Motivation and Problem Statement

Accurate CAC quantification requires models to handle complex vessel geometries and varying plaque morphologies. Current deep learning models struggle with:

- **Anatomical Ambiguity:** Misidentifying osseous structures (ribs/spine) as CAC due to similar Hounsfield Unit (HU) profiles.
- **Clinical Data Scarcity:** Rare but high-risk cases (e.g., Agatston > 1000) are severely underrepresented in public datasets like COCA.
- **Geometric Discontinuity:** Purely generative methods ignore the underlying fluid dynamics and Z-axis continuity of the coronary arteries.

3 Proposed Methodology

The **Physio-Twin** framework leverages a robust anatomical foundation to drive a physics-constrained generative process.

3.1 Phase 1: Validated Anatomical Scaffolding

The framework utilizes a **validated, failure-aware atlas registration framework** that achieved an **86.6% cohort-level success rate** and **82.75% mean localization accuracy** in preliminary benchmarks.

- **Geometric Foundation:** The validated Z-axis stability of the registration pipeline ensures continuity of coronary trajectories, critical for downstream hemodynamic modeling.

- **ROI-Aware Filtering:** Using heart-region masks to eliminate false positives, ensuring 95%+ of synthetic calcium remains within expected coronary territories as per project requirements.

Prior Work: Task 3 Implementation

3.2 Phase 2: Uncertainty-Aware Hemodynamic Surrogates

Building on prior work in Navier–Stokes-constrained PINNs for cerebrovascular flow, this phase develops a lightweight surrogate to estimate **Endothelial Shear Stress (ESS)**.

- **Physics Constraints:** The surrogate is regularized using a physics-residual loss (enforcing physically consistent flow behavior) inspired by the Navier–Stokes equations:

$$\mathcal{L}_{phys} = \|\rho(\mathbf{u} \cdot \nabla)\mathbf{u} + \nabla p - \mu \nabla^2 \mathbf{u}\|^2 + \|\nabla \cdot \mathbf{u}\|^2 \quad (1)$$

- **Feasibility:** The surrogate operates on reduced-resolution representations to ensure tractable training within the 350-hour timeline.
- **Uncertainty Handling:** Failure cases identified in distal vessel alignment (as documented in Task 3 analysis) will be explicitly handled by incorporating uncertainty-aware weighting.

3.3 Phase 3: Stochastic Plaque Growth (Core Innovation)

The central contribution is a **Latent SDE** model that simulates the deposition of calcium as a dynamic process:

$$dz = f(z, ESS) dt + \sigma(z) dW \quad (2)$$

- **Physiological Drift (f):** Guided by the ESS field, driving plaque toward low-stress and oscillatory flow zones.
- **Stochastic Perturbations (σ):** Captures biological variability. This explicitly leverages characterized failure modes from the registration pipeline (e.g., apical misalignment) to improve robustness in challenging anatomical cases.
- **Fallback Strategy:** A discretized stochastic growth process will be used if continuous SDE integration proves unstable.

3.4 Phase 4: Lightweight Physiological Texturing

A **lightweight texture synthesis module (diffusion or template-based)** will map the SDE-generated masks to realistic **Hounsfield Unit (HU)** distributions established in previous benchmarks (clipping to $[-200, 600]$). This work aligns with the broader **Hidden Fluid Mechanics (HFM)** objective by enabling data-driven inference of plaque–flow interactions.

4 Deliverables and Artifacts

- **Physio-Twin CLI Tool:** A modular Python pipeline for generating synthetic CAC scans, designed for seamless integration into the **PrediCT pipeline**.
- **Controllable Generation Engine:** Interface to parameters such as Target Agatston Score, Phenotype (Spotty/Dense), and Vessel Territory.

- **Scalable Synthetic Dataset:** A release of a **scalable synthetic dataset (target: 300–500 scans)**, including “long-tail” high-Agatston cases.
- **Benchmarking Report:** Statistical comparison between synthetic and real distributions (KS-test, EMD).

5 Evaluation and Validation

- **Structural Fidelity:** Morphological validation comparing lesion size, spatial clustering, and vessel-wise distribution against real CAC statistics.
- **Statistical Validation:** Comparing synthetic vs. real lesion counts using the **KS-test** or **Earth Mover’s Distance (EMD)**.
- **Ablation Study:** Comparing physics-guided vs. purely stochastic generation to quantify the impact of hemodynamic priors.
- **Expert Turing Test:** Visual assessment by **clinical experts** to evaluate realism (Target: < 60% distinguishability).

6 Timeline (350-Hour Track)

Week	Focus Area	Key Milestones
1–2	Integration	Coordinate mapping & registration scaffold setup
3–5	Hemodynamics	Training PINN surrogate with uncertainty weighting
6–8	Stochastic Growth	Implementing Latent SDE model & phenotypes
9–10	Texturing	Developing synthesis module & CLI interface
11–12	Validation	Expert review, statistical testing & final report

7 Execution Guarantee

Specialized background in **SDE-PDE coupled frameworks** and **biomechanical simulations** ensures the mathematical foundations of this project are feasible. **Given that the anatomical and validation backbone has already been implemented and benchmarked, this project focuses on extending an existing, validated system into a generative framework rather than building from scratch.** This framework serves as a foundation for future digital twin-based cardiovascular systems.

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